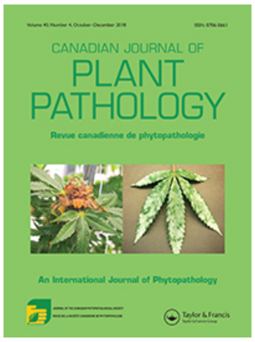

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Root and crown rot pathogens causing wilt symptoms on field-grown marijuana (*Cannabis sativa* L.) plants

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Abstract: Yellowing and wilting symptoms on field-grown *Cannabis sativa* (cannabis) plants followed by total plant collapse under conditions of extreme hot weather were observed in northern California in 2017. The crown regions of affected plants were dark and sunken and internal tissue discolouration extended 10–15 cm above the soil surface. Isolations made from the pith, vascular and cortical tissues in the crown region yielded *Fusarium oxysporum* (40% frequency), *F. brachygibbosum* (28% frequency), *Pythium aphanidermatum* (22% frequency), *Fusarium solani* and *F. equiseti* (5% frequency each). Pathogenicity tests were conducted on rooted plantlets to establish the extent of root and crown decay, as well as on mature stems to determine the extent of stem tissue colonization caused by these species. Extensive reduction in root length was caused by *F. solani*, *F. oxysporum*, *F. brachygibbosum* and *P. aphanidermatum* and wounding significantly enhanced disease development. Stem tissue colonization by these pathogens at wound sites was similarly extensive. Isolates of *F. equiseti* were non-pathogenic. Both *F. solani* and *P. aphanidermatum* caused plant mortality within 6–10 weeks following inoculation. In phylogenetic analyses using the internal transcribed spacer (ITS) rDNA region and the elongation factor 1 (*EF-1α*) region, *F. oxysporum* isolates from cannabis plants in northern California were grouped separately from all other *formae speciales* and from isolates previously recovered from British Columbia. Two isolates of *F. brachygibbosum* were identical to an isolate previously reported to infect almond stems in cold storage and field-grown seedlings in northern California. These findings indicate that a complex of pathogens potentially can cause root and crown rot under field conditions, resulting in wilt symptoms and collapse of cannabis plants.

Keywords: crown rot, fusarium root rot, pythium root rot, stem colonization, wilting

Résumé: En 2017, des symptômes du jaunissement et du flétrissement constatés sur le cannabis (*Cannabis sativa*) cultivé en champ, suivi de l'effondrement complet de la plante dans des conditions de temps extrêmement chaud, ont été observés dans le nord de la Californie. La région du collet des plants touchés était sombre et enfoncée, et la décoloration du tissu interne s'étendait de 10 à 15 cm au-dessus de la surface du sol. Les isolements faits à partir de la moelle ainsi que des tissus vasculaires et corticaux provenant de la région du collet ont produit *Fusarium oxysporum* (fréquence de 40%), *F. brachygibbosum* (fréquence de 28%), *Pythium aphanidermatum* (fréquence de 22%), *Fusarium solani* et *F. equiseti* (fréquence de 5% chacun). Des tests de pathogénicité ont été menés sur des plantules enracinées afin d'établir l'ampleur de la pourriture des racines et du collet causée par ces espèces, ainsi que sur des tiges matures pour déterminer l'importance de la colonisation des tissus de la tige. *F. solani*, *F. oxysporum*, *F. brachygibbosum* et *P. aphanidermatum* ont considérablement réduit la longueur des racines, et les lésions ont notablement accru le développement des maladies. La colonisation des tissus de la tige par ces agents pathogènes aux sites des lésions était tout aussi considérable. Les isolats de *F. equiseti* n'étaient pas pathogènes. *F. solani* et *P. aphanidermatum* ont causé la mort des plants en 6 à 10 semaines suivant l'inoculation. Dans le cadre des analyses phylogénétiques basées sur la région de l'espaceur transcrit interne (ITS) de l'ADNr et la région du facteur d'élongation 1 (*EF-1α*), les isolats de *F. oxysporum* provenant de plants de cannabis du nord de la Californie ont été regroupés séparément de toutes les autres *formae speciales*, ainsi que des isolats préalablement récupérés de Colombie-Britannique. Deux isolats de *F. brachygibbosum* étaient identiques à un isolat qui, auparavant,

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avait infecté des tiges d'amandier entreposées en chambre froide et des semis cultivés en champ dans le nord de la Californie. Ces résultats indiquent qu'un complexe d'agents pathogènes peut causer la pourriture des racines et du collet dans des conditions réelles, provoquant des symptômes de flétrissement et d'effondrement chez les plants de cannabis.

Mots clés: colonisation de la tige, flétrissement, pourridié pythien, pourriture du collet, pourridié fusarien

Introduction

Cannabis sativa L., a member of the family Cannabeaceae, is grown as a source of fibre and seed (as industrial hemp) or for medicinal and recreational use (as marijuana or cannabis) in Canada and the USA. For commercial production in Canada, hemp is grown outdoors (Laate, 2012) while cannabis is presently grown primarily indoors under greenhouse conditions or in controlled environment rooms (Punja & Rodriguez, 2018) by licensed producers. Cannabis production in the USA, in states where it has been approved by the state legislature for the purposes of medicinal or recreational use, occurs both indoors and outdoors. As of January 2018, nine USA states allow the legal sale and possession of marijuana for both medical and recreational use, while Vermont and the District of Columbia have legalized personal use but not commercial sale (Anonymous, 2017). In addition, 31 US states and the District of Columbia have passed state laws allowing some degree of medical use of marijuana. Legislative approval for recreational use of cannabis in Canada has been granted and policies will be enforced in all provinces in October 2018.

California is the largest producer in the USA of both medical and recreational marijuana, with both indoor and outdoor production (Gettman, 2006). In August–September 2017, cannabis plants cultivated by a licensed producer in a field located in northern California displayed symptoms of yellowing and wilting in comparison to unaffected plants (Fig. 1). During conditions of extreme hot weather (exceeding 35°C during the daytime for several weeks during July–September), plants had completely collapsed (Fig. 1). Previously reported pathogens that could potentially cause wilting and collapse of *C. sativa* plants grown outdoors include *Fusarium oxysporum* f. sp. *cannabis* and *F.o.* f. sp. *vasinfectum* (McPartland, 1996, 2004). A recent report showed that *Pythium aphanidermatum* (Edson) Fitzp. also caused stunting and wilting and occasional death of industrial hemp plants (Beckerman et al., 2017), and has been recovered also from greenhouse-grown rooted cannabis cuttings and flowering plants (Punja & Rodriguez, 2018). In indoor hydroponic cultivation, *F. oxysporum* and *F. solani* were shown to cause root rot and wilt of cannabis plants (Punja & Rodriguez, 2018).

The objective of this study was to isolate, characterize and identify the potential causal agent(s) associated with the wilting symptoms on the affected cannabis plants in outdoor production, and to confirm pathogenicity of any recovered isolates.

Materials and methods

Plant samples

A number of affected cannabis plants showing symptoms of yellowing, wilting and collapse (Fig. 1) were sampled from a production field in northern California in September 2017. The cannabis strains observed to be most affected by crown and root disease in this field were 'Agent Orange' and 'Maui Dawg' (A. Abare, personal communication) and represented about 4–5% of the total plant population. The plants had been initiated from stem cuttings grown in 4 L pots in an outdoor nursery and then transplanted into raised fabric pots (Grassroots; www.cannabisfabricpots.com) which contained a mixture of pasteurized soil, vermiculite, vermicomposted plant materials and a blend of fertilizer and were provided with drip irrigation. From a total of six symptomatic plants, approximately 20 tissue samples were collected. Each tissue sample was comprised of either outer bark and cortical tissues taken from sunken dark areas near the crown of the plants (10 samples) (Fig. 2) or internal discoloured areas extending 10–15 cm upward from the crown that included the pith and vascular tissues (10 samples). Samples were placed in individual sealed plastic bags and stored at 4°C for 7–10 days until isolations were made. Small segments of tissue (5 mm^2) were cut from the samples, dipped into a 0.5% NaOCl solution for 30 s followed by 20 s in 70% EtOH, rinsed thrice in sterile water, and blotted dry on sterile paper towels. The tissue pieces were cut into smaller pieces (2–3 mm^2) and plated onto potato dextrose agar (PDA, Sigma Chemicals, St Louis, MO) amended with 100 mg L^{-1} streptomycin sulphate and incubated at ambient temperature (23 ± 2°C) for 5–7 days. Subcultures were made from the growing margins of colonies and transferred to fresh PDA for subsequent identification to genus level using morphological criteria (Van der Plaats-Niterink, 1981; Leslie &

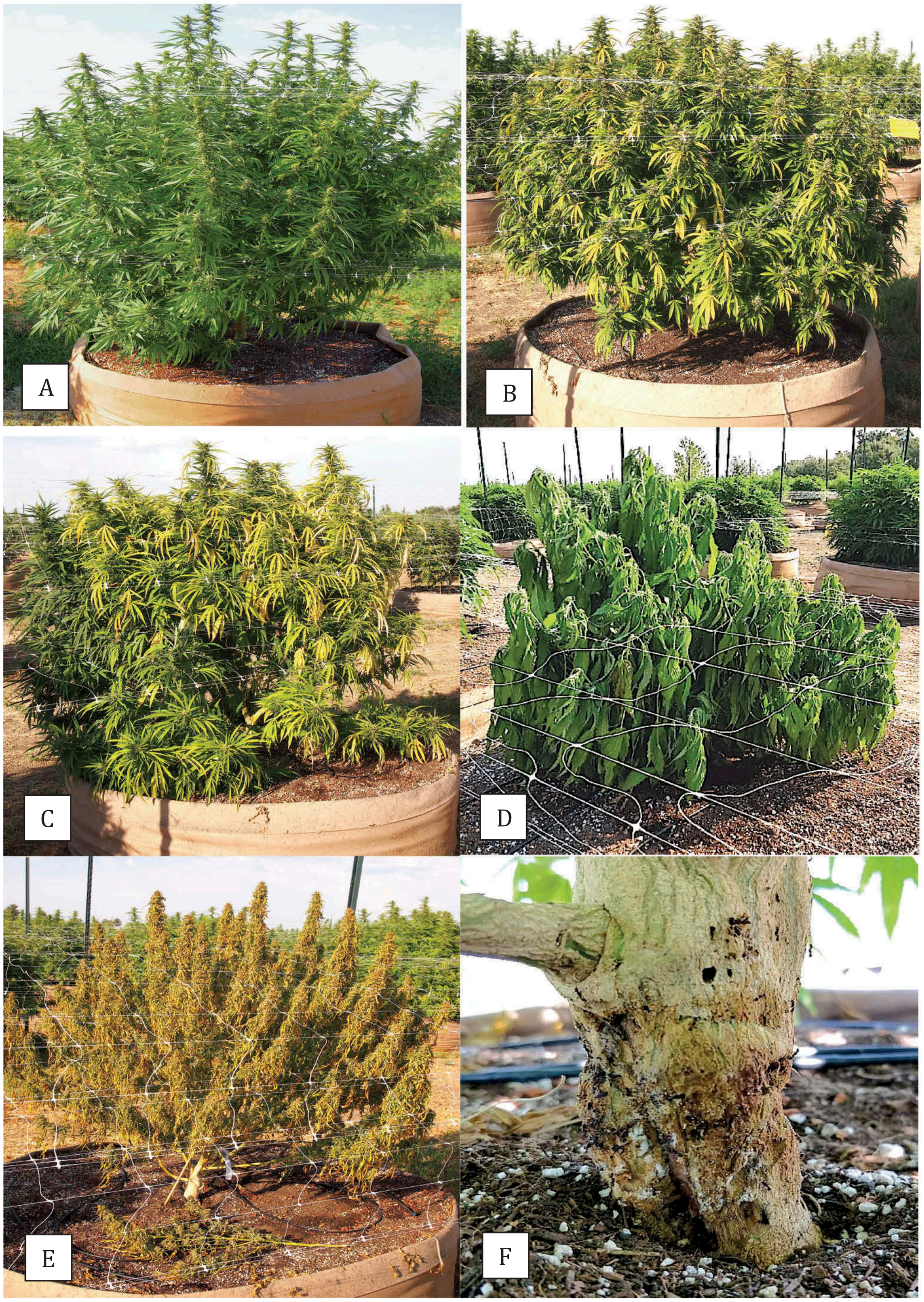

Fig. 1 (Colour online) Symptoms of root and crown rot on field-grown cannabis plants. (A) A healthy cannabis plant of approximately 16 weeks of age in full flower; (B) Early symptoms of general yellowing of the foliage on a diseased plant; (C) Pronounced yellowing of a diseased plant; (D) Rapid wilting of diseased plant under hot weather conditions; (E) Death and desiccation of diseased plant; (F) Dark sunken area at the crown extending up to 15 cm from the soil surface.

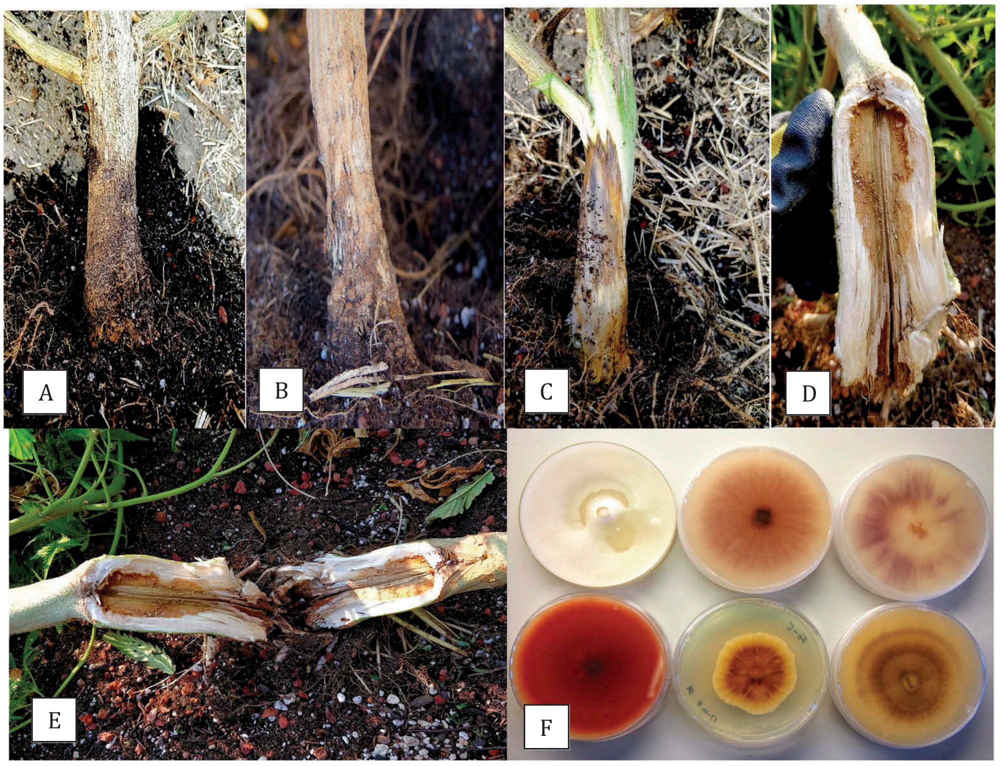

Fig. 2 (Colour online) Symptoms of root and crown rot on field-grown cannabis plants. (A) Crown region of diseased plant showing a darkening of the underlying tissues extending into the roots below the soil surface; (B) Superficial mycelial growth on sunken areas at the crown; (C, D) Dark staining and rot of the underlying tissues progressing upwards from the crown; (E) Reddish-brown internal decay originating from the crown and root zone and extending upwards; (F) Colonies of *Pythium* and *Fusarium* originating from diseased tissues – top row: *P. aphanidermatum* (far left) and *Fusarium oxysporum* CA-1 and CA-2 (middle and right Petri dishes); bottom row: *F. brachygibbosum* (far left), *F. equiseti* (middle dish), *F. solani* (far right). Isolates of these species (except *F. equiseti*) originated from outer bark, cortical and vascular tissues; isolates of *F. equiseti* originated from the bark only.

Summerell, 2006) Hyphal tip sub-cultures were made of representative isolates (a total of 40) and stored on PDA slants at 4°C for subsequent identification to species level using PCR.

Identification of isolates by PCR

Cultures tentatively identified using morphological criteria (Van der Plaats-Niterink, 1981; Leslie & Summerell, 2006) as *Pythium* and *Fusarium* spp. that originated from diseased plants were sent to the University of Guelph Laboratory Services, Agriculture and Food Laboratory, Guelph, ON (www.guelphlabservices.com) for species identification by PCR using the primers ITS1–ITS4 (ITS1-F CTTGGTCATTTAGAGGAAGTAA and ITS4 TCCTCCGCTTATTGATATGC). In addition, for *Fusarium* spp., the elongation factor 1α (*EF*-1 α) primer set EF-1 (5′ ATG GGT AAG GAG GAC AAG AC 3′) and EF-2 (5′ GGA GGT ACC AGT GAT CAT GTT 3′) (O'Donnell et al., 1998) was also used according to the following protocol. Cultures of representative isolates from cannabis tissues were grown in potato dextrose broth at room temperature for 7 days and DNA was extracted from harvested mycelium using the QIAGEN DNeasy Plant Mini Kit. Aliquots of 1 uL containing 5–20 ng DNA were used for PCR in a 25 uL reaction volume consisting of 2.5 uL 10× buffer (containing 15 mM $MgCl_2$), 0.5 uL 10 mM dNTP, 0.25 uL Taq DNA Polymerase (QIAGEN), 0.25 uL 10 mM forward and reverse primers, as well as 20.25 uL DNAse- and RNAse-free water (Invitrogen). All PCR amplifications were performed in a MyCycler thermocycler (BIORAD) with the following program: 3 min at 94°C; 30 s at 94°C, 30 s at 60°C, 3 min at 72°C (35 cycles); and 7 min at 72°C. PCR products were separated on 1% agarose gels and bands of the expected size (~700 bp) were purified with QIAquick Gel Extraction Kit and sent to Eurofins Genomics (Eurofins MWG Operon LLC 2016, Louisville, KY) for sequencing. The ITS1-5.8S-ITS2 or EF-1α sequences were compared to the corresponding sequences from the National Center for Biotechnology

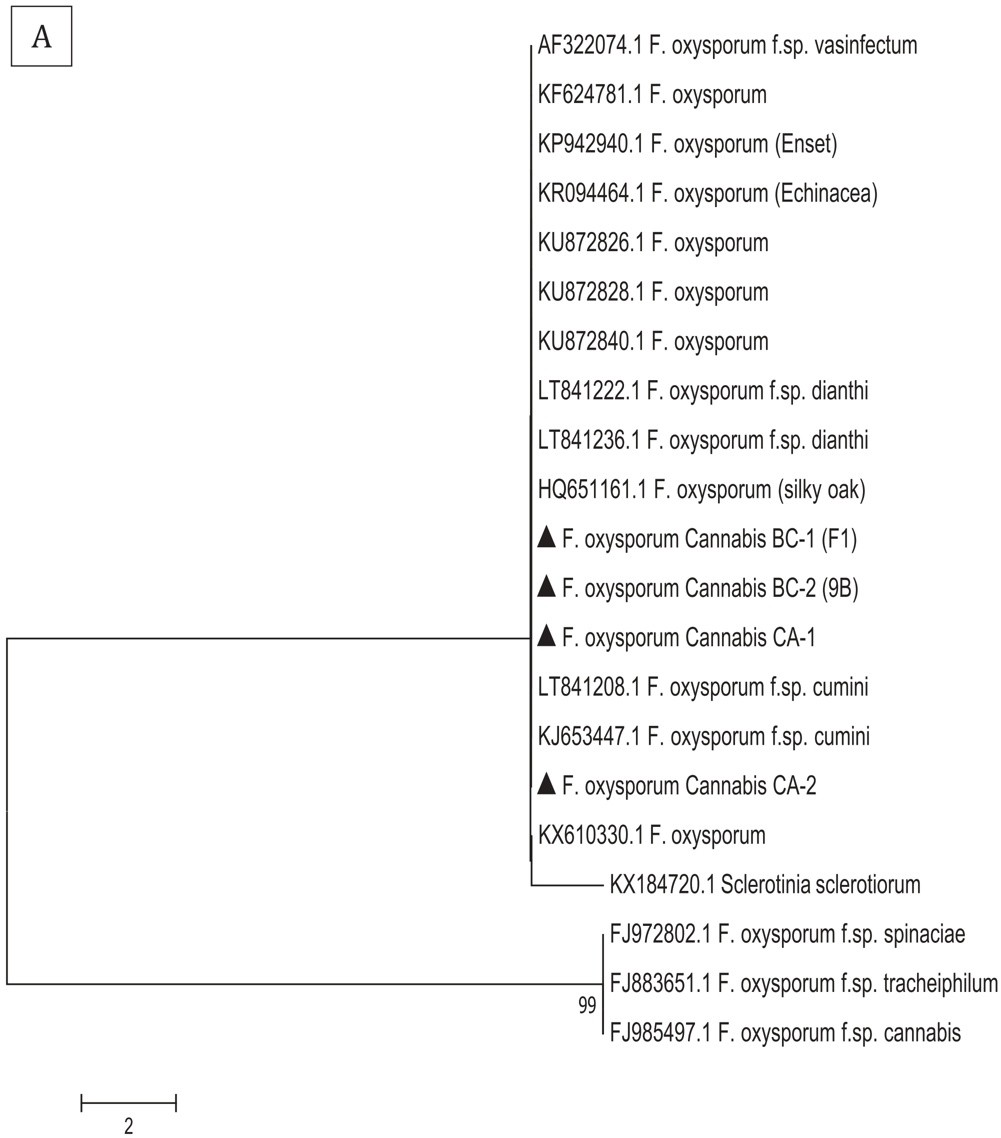

Fig. 3 Phylogenetic tree of *Fusarium oxysporum* isolates originating from cannabis plants using (A) ITS1-5.8S-ITS2 sequences and (B) EF-1α sequences compared to isolates from a range of other hosts (GenBank numbers are shown). A bootstrap consensus tree was inferred from 1000 replicates to represent the distance using the neighbour-joining (NJ) method. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates were collapsed. Isolates BC-1 and BC-2 were from British Columbia obtained in a previous study (Punja & Rodriguez, 2018) and CA-1 and CA-2 were from California (present study). The scale bar indicates the expected number of nucleotide substitutions.

Information (NCBI) GenBank database. Multiple sequence alignment of the respective isolates of each species was done using the CLUSTAL W program (http://www.genome.jp/tools/clustalw). The consensus sequences of *Fusarium* species were subsequently included in a phylogenetic analysis using the neighbour-joining (NJ) method (Saitou & Nei, 1987; Tamura et al., 2004) in the software MEGA v. 5 (Tamura et al., 2011). A bootstrap consensus tree was inferred from 1000 replicates. *Sclerotinia sclerotiorum* was used as an outgroup. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates were collapsed (Felsenstein, 1985).

Pathogenicity tests

Rooted cuttings. Stem cuttings (~15 cm in length) of cannabis strain 'CBD Therapy' were rooted by inserting them into 5 cm^2 rockwool plugs that had been pre-soaked in hydroponic nutrient solution (Current Culture H_2O®) (https://www.cch2o.com/). The cuttings were misted with a

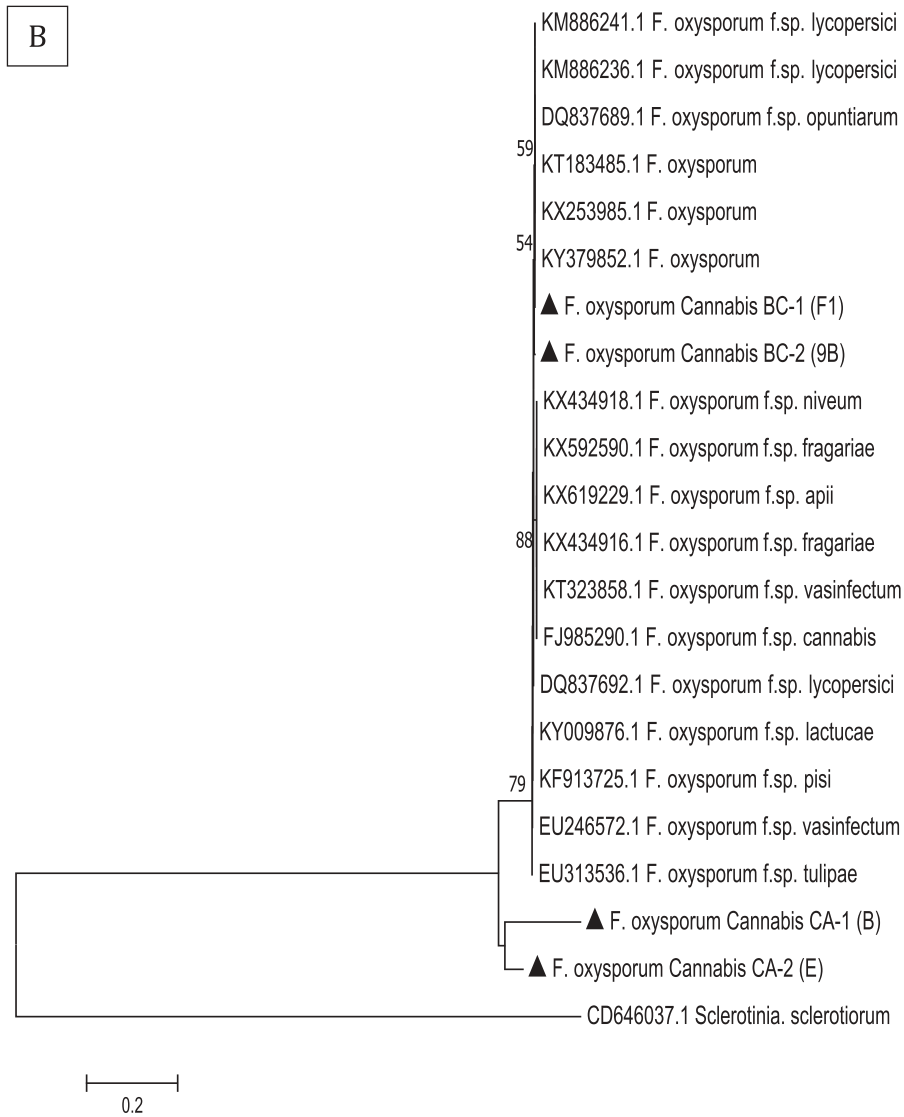

Fig. 3 Continued.

hand atomizer and covered with a plastic dome to maintain high humidity and placed on the laboratory bench at ambient room temperature (22–25°C) with a 16-h photoperiod. Misting was repeated every other day. Once rooted (about 2 weeks later), the rockwool plugs were transferred to 8 cm^2 pots containing a 1:4 mix of vermiculite:coconut fibre (Canna Coco brick) (http://www.cannagardening.ca/coco_brick) and placed in a plastic tray containing tap water. The plants were grown under Sunblaster brand CFL lamps (200 watts, 24 h photoperiod, light intensity of 240 umol m^{-2} s^{-1}), temperature range of 21–25°C. After 2 weeks, the plants were inoculated with two isolates of each of the previously identified *Pythium* or *Fusarium* species. The inoculum was prepared by transferring a mycelial plug into 75 mL of potato dextrose broth in 125 mL Erlenmeyer flasks and incubating on a shaker at 100 rpm for 7 days under ambient conditions. The contents were transferred to 150 mL of water and blended in a Waring blender at high speed for 10 s. The suspension, which consisted of mycelium of *Pythium* or *Fusarium*, as well as spores in the case of *Fusarium*, was serially diluted (in 10-fold increments) and 0.5 mL aliquots were spread onto a series of Petri dishes containing PDA and placed at 23–25°C for 5 days to estimate colony-forming units per mL. A volume of 20 mL of inoculum of each isolate was poured into the pot containing the plant, and a scalpel was inserted vertically at the base of the plant into the soil at four locations to create wounds. Control plants were similarly wounded and received 20 mL of sterile water. Non-

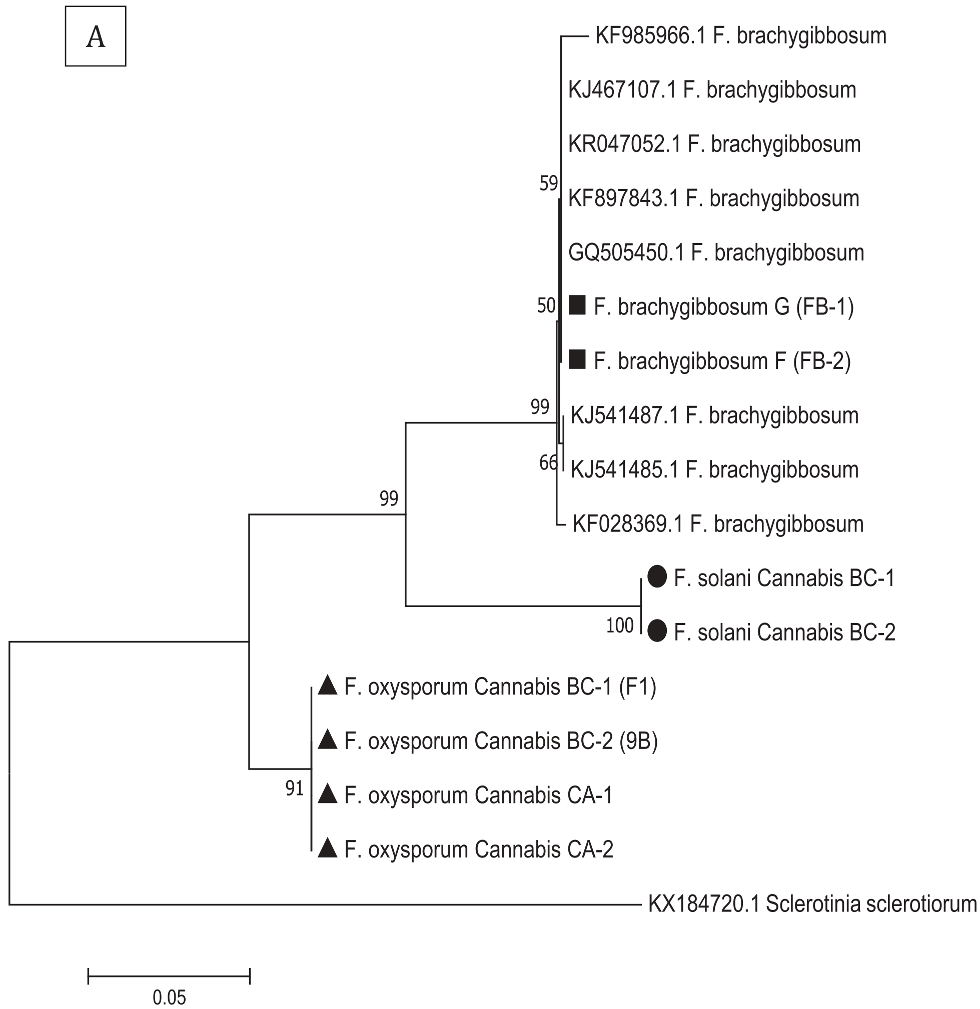

Fig. 4 Phylogenetic tree of *F. brachygibbosum* isolates originating from cannabis plants using (A) ITS1-5.8S-ITS2 sequences and (B) EF-1α sequences compared to isolates from a range of other hosts (GenBank numbers are shown). A bootstrap consensus tree was inferred from 1000 replicates using the neighbour-joining (NJ) method. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates were collapsed. Isolates BC-1 and BC-2 of *F. oxysporum* and *F. solani* were from British Columbia obtained in a previous study (Punja & Rodriguez, 2018) and isolates FB-1 and FB-2 of *F. brachygibbosum* and *F. oxysporum* CA-1 and CA-2 were from California (present study). The *F. brachygibbosum* isolates were grouped with a canker-causing isolate from almond seedlings in California (GenBank KX421379.1). The scale bar indicates the expected number of nucleotide substitutions.

wounded inoculated plants were also included. There were four replicate plants per isolate. After 5 weeks, all plants were examined for foliar symptoms (leaf chlorosis or necrosis, stunted growth) and depotted, and the extent of root browning and root length was compared to the non-inoculated control plants. The root length measurements were expressed as means ± standard errors. Root, crown and lower stem segments ~1 cm long were surface-sterilized for 30 s in a 0.5% NaOCl solution, rinsed three times for 30 s in sterile water, blotted dry on a sterile paper towel and then plated on PDA. Plates were incubated at room temperature (22 ± 3°C) for three days. The experiment was conducted twice.

Stem tissues. Mature cannabis plants (12–14 weeks old) were harvested and side branches (3–6 mm in diameter) were cut into sections 15 cm in length and brought back to the laboratory. The surfaces of the stem and cut edges were wiped with tissue paper dipped in a 70% EtOH solution and placed inside a plastic box. Using a scalpel, a 6 cm long strip of the stem surface (including the bark and epidermal tissues) was removed, or a 0.3 × 0.3 cm piece was cut and removed to expose the underlying tissue. The wounded stems were inoculated with a 0.5 cm diameter mycelial plug taken from a 7 day-old PDA

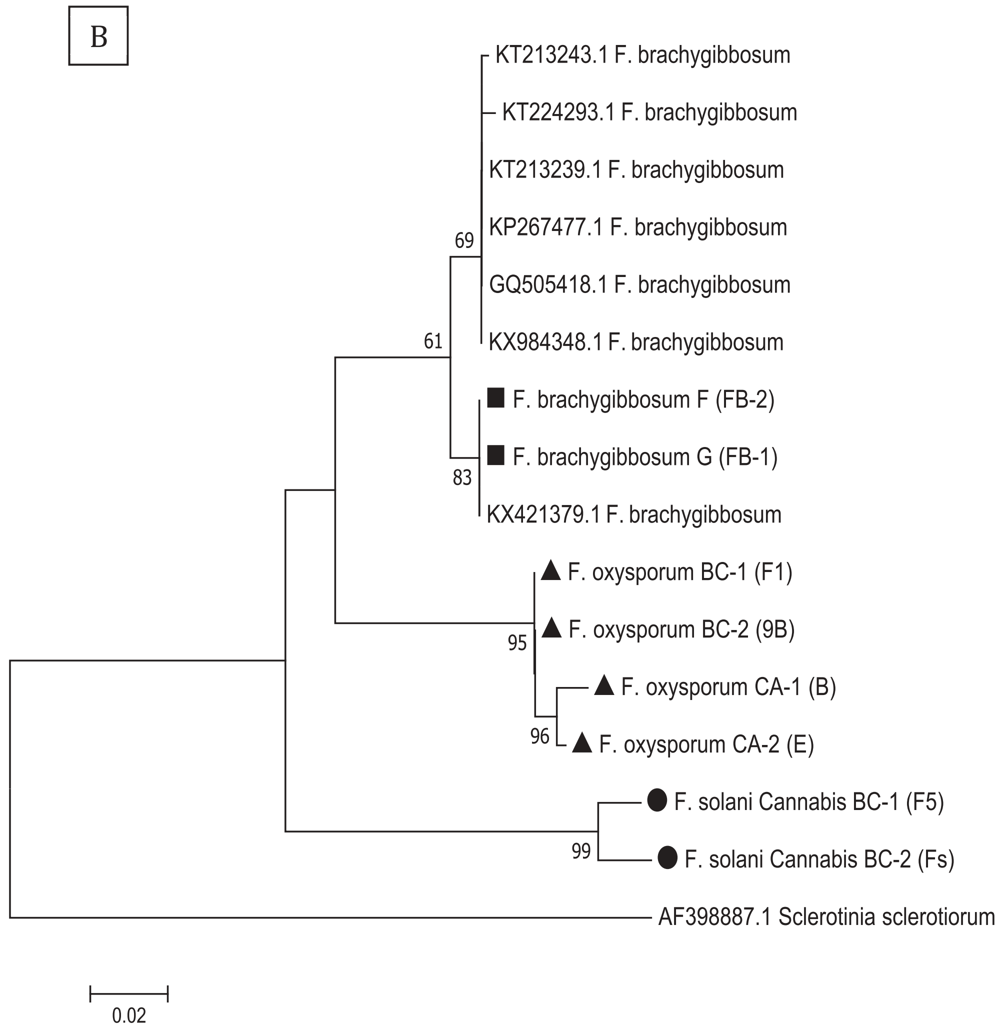

Fig. 4 Continued.

culture of the isolate to be tested, and placed mycelial side down on the exposed tissues. The stems were carefully transferred into test-tubes (150 × 20 mm) containing a moistened cotton plug placed on the bottom and sealed with a cap and stored horizontally in a test-tube rack at ambient conditions (temperature range 22–25°C). Controls included non-inoculated stems and inoculated non-wounded stems. The extent of browning of the stems and colonization by mycelial growth extending from the plug along the stem surface was rated on a scale of 0 to 4, where 0 = no growth from plug, no browning; 1 = growth covering up to 25% of the stem surface, mild browning; 2 = growth extending up to 50% of the stem surface, visible browning; 3 = growth extending up to 75% of the stem surface, extensive browning; and 4 = growth covering the entire stem segment, severe browning. There were five replicate stems per isolate and the experiment was conducted twice. Rating measurements were expressed as means ± standard errors.

Results and discussion

Field symptoms

At the time of sampling, cannabis plants had reached a height of 1.2–1.6 m and were 1.8–2 m wide and in full flower (Fig. 1*a*) after 16 weeks of growth. Diseased plants exhibited a general yellowing of the foliage (Fig. 1*b*), or in some cases showed more pronounced yellowing on one side of the plant (Fig. 1*c*). Some plants showed a complete wilting that progressed within 48–72 h (Fig. 1*d*) under extremely hot conditions (daytime temperatures exceeding 35°C). In many cases, these plants died rapidly and became desiccated (Fig. 1*e*). Four of the six severely affected plants had dark sunken areas at the crown that extended upward 15 cm from the soil surface (Fig. 1*f*). A closer examination of the crown region showed the darkening of the tissues extended into the roots below the soil surface (Fig. 2*a*) and many of the roots were necrotic. Mycelial growth could be observed in the sunken areas at the crown (Fig. 2*b*). Removal of the bark revealed a dark staining and rot of the underlying tissues that progressed upwards from the crown (Fig. 2*c,d*). When the stem was split, the reddish-brown internal decay could be seen originating from the crown and root zone and extended upwards (Fig. 2*e*). The symptoms observed are characteristic of a crown rot disease as they were not limited to the vascular tissues.

Identification of cultures

From around 120 plated tissue segments originating from six diseased plants, each one produced colonies that were identified as *Fusarium* or *Pythium* species based on morphological characteristics of these genera (Fig. 2*f*). The *Pythium* colonies were fast-growing (covering a 9 cm diameter Petri dish in 2–3 days) and two isolates 2–6 and 3–2 (GenBank accession nos. MH789989 and MH789990) were identified as *P. aphanidermatum* (100% identity with GenBank sequences KU211462.1 from soybean (Rojas et al., 2017) and AY598622.2 from CBS118.80 (Levesque & de Cock, 2004)). The frequency of recovery was 22% and all isolates originated from the bark or cortical tissues taken from dark sunken regions at the crown. The remaining 78% of isolates were identified as *Fusarium oxysporum* (40% frequency), *F. brachygibbosum* (28% frequency), *Fusarium solani* and *F. equiseti* (5% frequency each) (Fig. 2*f*). Isolates of these species (except *F. equiseti*) originated from outer bark, cortical and vascular tissues; isolates of *F. equiseti* originated from the bark only.

Phylogenetic analysis of two isolates of *F. oxysporum* (CA-1, CA-2) using the ITS and EF-1α regions placed them within a clade that contained isolates of different *formae specialis* from a range of hosts (Fig. 3*a,b*), including two isolates of *F. oxysporum* from British Columbia (BC-1, BC-2) that were recovered from the roots of hydroponically grown cannabis plants (Punja & Rodriguez, 2018). The two isolates of *F. brachygibbosum* (F, G) were grouped with a range of isolates of this species from other hosts and sources, and were distinct from isolates of *F. oxysporum* and *F. solani* (Fig. 4*a,b*). There was 100% sequence identity in the EF-1α region with an isolate of *F. brachygibbosum* causing a canker disease on almond seedlings in California (GenBank KX421379.1) (Stack et al., 2017). All isolates of *F. brachygibbosum* from cannabis produced distinctly red pigmented colonies on PDA (Fig. 2*f*). Sequences of *F. oxysporum* isolates CA-1 and CA-2 have been deposited in GenBank (accession nos. MH789985 and MH789986 for ITS and MH844832 and MH844833 for EF-1α) and *F. brachygibbosum* isolates F and G (accession nos. MH789987 and MH789988 for ITS and MH844834 and MH844836 for EF-1α).

Pathogenicity tests

For the inoculation experiments on rooted cannabis plants, two representative isolates each of *F. oxysporum*, *F. brachygibbosum*, *F. solani*, *P. aphanidermatum* and *F. equiseti* were included; the inoculum levels ranged from 1×10^2 to 1×10^3 CFU per mL for *P. aphanidermatum* and 1×10^4 to 1×10^5 for the *Fusarium* species. Isolates of *F. oxysporum*, *F. brachygibbosum* and *P. aphanidermatum* caused similar reductions in root length compared with the non-inoculated control plants (Fig. 5*a*). Only plants that were wounded prior to inoculation showed a reduction in root growth; non-wounded inoculated plants did not develop symptoms (data not shown). Inoculation with *F. solani* (isolates Fs, F5) caused the greatest reduction in root length, confirming the observations from a recent study on the pathogenicity of this species on cannabis plants (Punja & Rodriguez, 2018). While *F. equiseti* was not pathogenic on plants (data not shown), this species has also been recovered from cannabis flower buds and was shown to be weakly pathogenic on flowers compared with the highly aggressive *F. solani* (Punja, 2018). Plants inoculated with *F. oxysporum* and *F. solani* developed symptoms of stunting and yellowing (Fig. 6*a*) while isolates of *F. brachygibbosum* and *P. aphanidermatum* caused stunting of plants within 5 weeks following inoculation (Fig. 6*b*). Total collapse and death of some plants was observed following inoculation with *P. aphanidermatum* (Fig. 6*c*). Browning of the pith tissues was observed following inoculation with *F. oxysporum* isolate CA-2 (Fig. 6*d*) which developed within 2–3 weeks after inoculation and progressed upwards from the crown region within 5 weeks (Fig. 6*e*). A general discolouration of the bark and inner cortical tissues was observed following inoculation with *P. aphanidermatum* (Fig. 6*f*).

The recovery of four pathogenic species from diseased crown and vascular tissues of cannabis plants makes the establishment of the importance of each individual

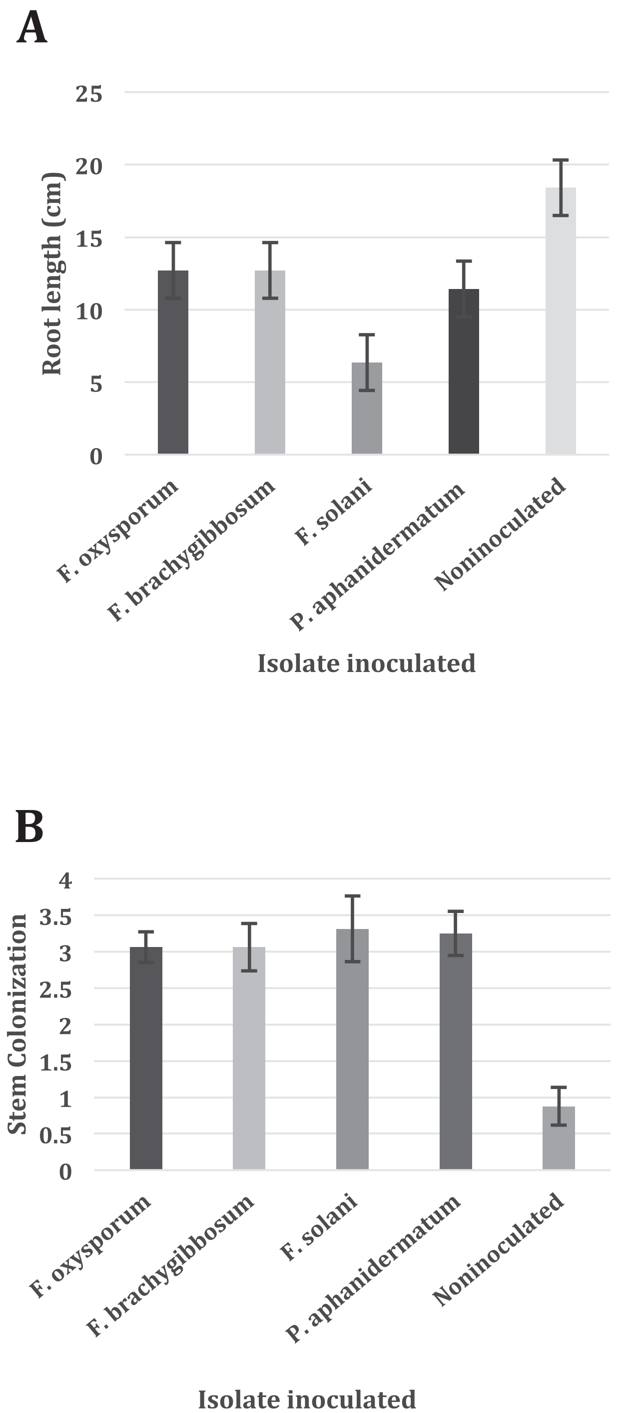

Fig. 5 Effect of inoculation of cannabis plants and stem tissues with *F. oxysporum, F. brachygibbosum, F. solani* and *P. aphanidermatum*. (A) Reduction in root length compared with the non-inoculated control. Data are from four replicate plants; (B) Stem colonization by mycelium. Data are from 5 replicate stems. Both experiments were conducted twice. Data are the means ± standard error.

species difficult to discern. Species of *Fusarium* and *Pythium* were reported to infect roots of hydroponically grown cannabis plants (Punja & Rodriguez, 2018) but this is the first report showing *F. brachygibbosum* as a pathogen on cannabis. Wounding was required for infection by all isolates tested – such wounds can develop during transplanting in the field or from rodent damage to the root system that had occurred on many plants in this particular field earlier in the growing season (A. Abare, personal communication).

Inoculation of stem tissues showed that all pathogens were capable of causing extensive browning (Fig. 7*a*), accompanied by mycelial colonization (Figs 5*b* and 7*b*), while profuse sporulation was observed on the stem surface by the *Fusarium* species (Fig. 7*c,d,e*). Wounded control stems had only mild browning. The aerial release of spores from colonized stem tissues of diseased cannabis plants can result in spread of inoculum to adjacent plants in the field, as well as onto flower buds (Punja, 2018). Spread of *F. oxysporum* from diseased plant tissues has been reported for a number of hosts (Rowe et al., 1977; Gamliel et al., 1996; Katan et al., 1997; Rekah et al., 2000; Scarlett et al., 2015). Additional inoculum sources may be from the planting medium or from soil and diseased plants in adjacent fields. The isolates of *F. brachygibbosum* (F, G) from cannabis plants showed identical ITS sequences to those recovered previously from propagated almond trees in northern California (Stack et al., 2017). The symptoms on almond included cankers on bare-root trees in cold storage or in the field soon after planting, which resulted in death of numerous trees. The pathogen was also recovered from asymptomatic tissues, soil, adjacent wheat rotation crop and the surface of field equipment (Stack et al., 2017) and is believed to be widespread in California soils and is likely an opportunistic pathogen (Chitambar, 2016). *Fusarium brachygibbosum* has also been reported to cause a stalk rot of corn in China (Shan et al., 2017), characterized by a dark brown disintegration of internal stalk tissues. Watermelon wilt and plant death due to *F. brachygibbosum* was also reported from Sonora, Mexico, which caused a cortical rot on the upper portion of the taproot (Renteria-Martinez et al., 2015). Other plant species on which stem die-back and wilt were reported to be caused by *F. brachygibbosum* include olive in Tunisia (Trabelsi et al., 2018) and Euphorbia in Oman (Al-Mahmooli et al., 2013). The symptoms observed on cannabis plants i.e. wilting, die-

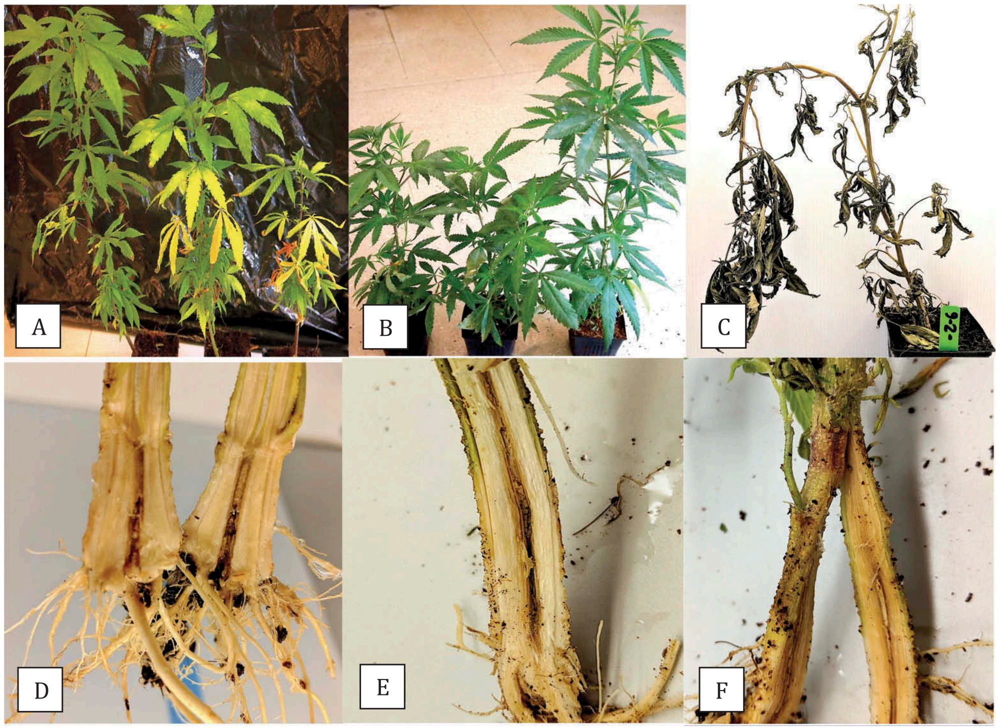

Fig. 6 (Colour online) Pathogenicity studies conducted on rooted cannabis cuttings using isolates of *Fusarium* and *Pythium* species. (A) *F. oxysporum* (middle plant) and *F. solani* (right plant) compared with control plant (far left). Photo was taken 5 weeks after inoculation and shows yellowing and stunting symptoms; (B) Non-inoculated plant (far right) compared with *F. brachygibbosum*-inoculated (middle) and *P. aphanidermatum*-inoculated plant (far left). Inoculated plants were stunted after 5 weeks; (C) Plant death following inoculation with *P. aphanidermatum*; (D) Internal pith discolouration progressing upwards from the crown region in a *F. oxysporum*-inoculated (isolate CA-2) plant; H) Advanced pith and xylem discolouration 5 weeks following *F. oxysporum* inoculation; (I) General discolouration of the bark and inner cortical tissues following inoculation with *P. aphanidermatum.*

back, root and crown rot, are consistent with the reports of *F. brachygibbosum* symptom development on other host plants. In addition, on young inoculated plants, disease symptoms include stunting and yellowing of lower leaves.

While *F. oxysporum* was the most frequently isolated pathogen in this study (40%), the recovery of *F. brachygibbosum* (28% frequency) and *P. aphanidermatum* (22% frequency) and their pathogenicity on cannabis plants suggest they likely have a role in the disease complex. Simultaneous co-inoculations of multiple pathogens were not performed in this study but may have additive effects in promoting disease symptoms. On olive and watermelon plants, other species of *Fusarium*, including *F. solani* and *F. oxysporum*, were also associated with diseased tissues that contained *F. brachygibbosum*, as part of a disease complex (Renteria-Martinez; Trabelsi et al., 2017). On almond, both *F. acuminatum* and *F. avenaceum* were also found to be present in diseased tissues (Marek et al., 2013). Both *F. oxysporum* and *F. solani* have been reported to cause root and crown rot on a number of different hosts worldwide (Nagao et al., 1994; Ozbay & Newman, 2004; Golzar et al., 2007; Punja & Parker, 2009; Moya et al., 2011; Elmer, 2015), while *P. aphanidermatum* causes root and crown rot on many species as well, especially at temperatures of 30–35°C (Kucharek & Mitchell, 2000; Sutton et al., 2006). On industrial hemp plants, *P. aphanidermatum* was recently reported to cause stunting and wilting, with rotted roots and crown lesions (Beckerman et al., 2017; Schoener et al., 2017), and was previously reported to cause seedling damping-off (McPartland, 1996). Collapse of some plants following inoculation with *P. aphanidermatum* was observed in the present study, as reported by Beckerman et al. (2017). Collapse and death of hydroponically grown cannabis plants under greenhouse conditions due to *P. aphanidermatum* has also recently been reported from British Columbia (Punja & Rodriguez, 2018).

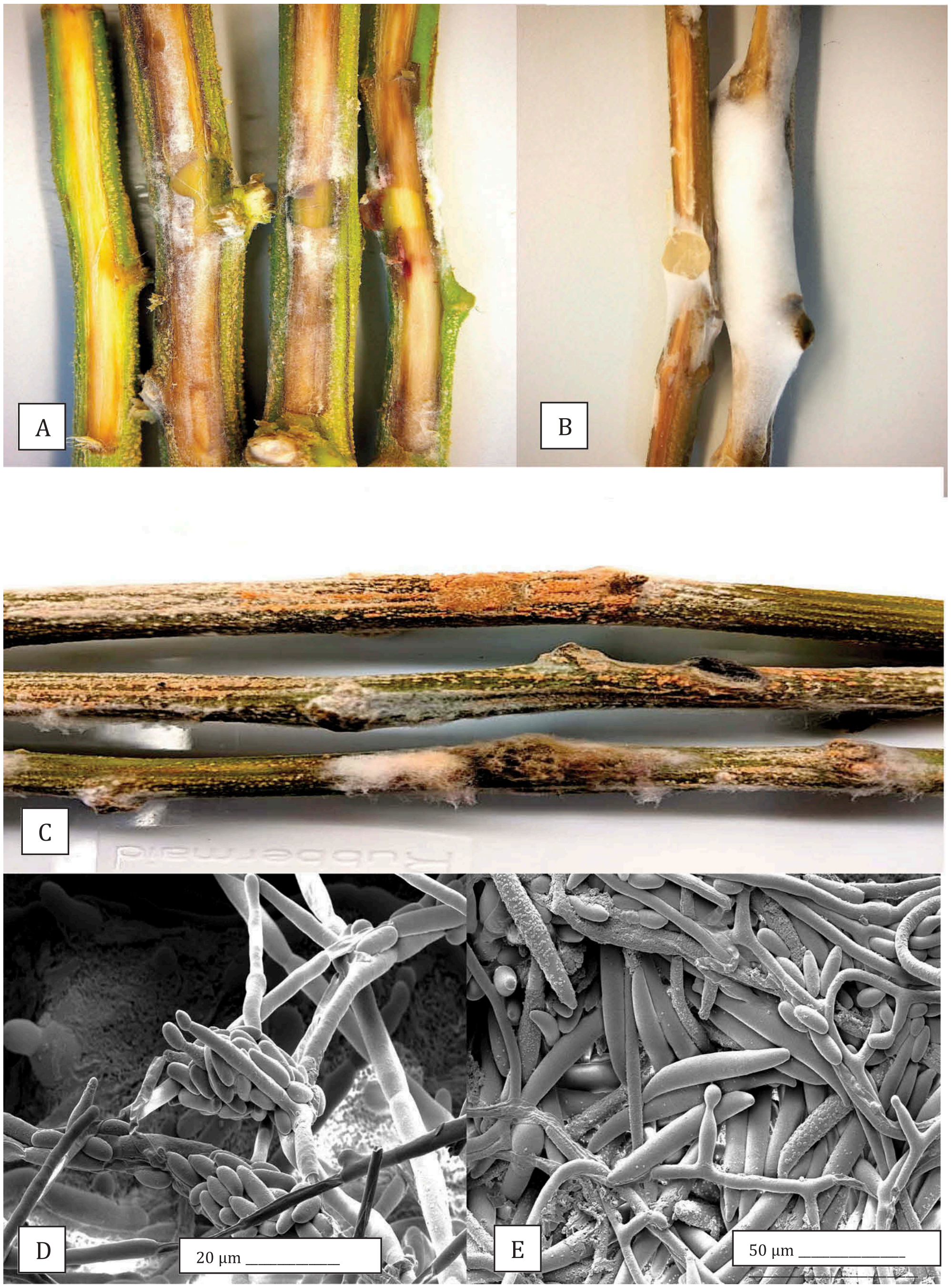

Fig. 7 (Colour online) Inoculation of wounded cannabis stem tissues with *Fusarium* and *Pythium* species. (A) Internal browning accompanied by mycelial colonization by *Fusarium* species after 7 days. Inoculations made were (left to right): control (wounded only), *F. oxysporum, F. solani, F. brachygibbosum*; (B) Inoculation with *P. aphanidermatum* showing internal browning (left) and external mycelial colonization (right); (C) Profuse sporulation on the surface of the bark of stems following inoculation with *F. oxysporum* (top), *F. solani* (middle) and *F. brachygibbosum* (bottom). Photographs were taken 7 days after inoculation with mycelial plugs that were placed on the surface of the wounded stems; plugs were removed prior to photographs being taken; (D, E) Scanning electron microscopy images of profuse sporulation of *F. oxysporum* on cannabis stem surfaces.

The presence of multiple species of *Fusarium*, in addition to *P. aphanidermatum*, on cannabis plants, is likely to exacerbate the disease symptoms and contribute to a rapid decline and death of infected plants, especially under hot weather or water-stressed conditions. Predisposition of cannabis plants during periods of excessive heat or drought is possibly a contributing factor in promoting disease. On almond, cold-stored nursery stock with cryptic infection by *F. brachygibbosum* developed disease under conditions of predisposing physiological stress (Stack et al., 2017). Our observations suggest that cannabis strains can differ in susceptibility to root and crown rot although there have been no previous efforts to screen different strains for potential tolerance to disease. Management of root and crown rot of cannabis will necessitate an integrated approach that will require knowledge of the source(s) of inoculum, understanding the role of predisposing stress factors, and the effects of soil-associated factors such as moisture levels, microbial activity and nutrient levels, on disease development.

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